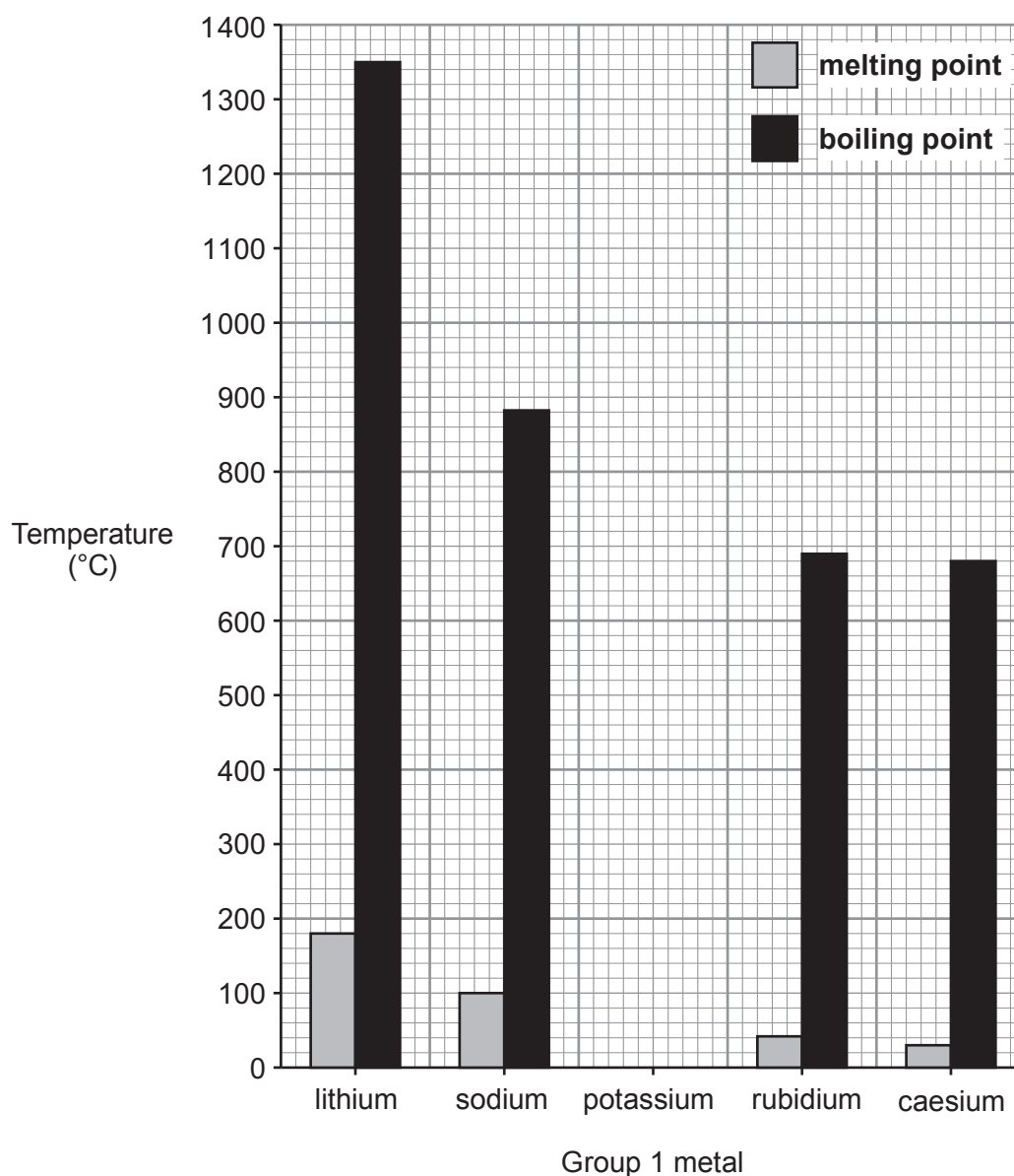


3. (a) The chart below shows the melting points and boiling points of some Group 1 metals. The data for potassium is missing.



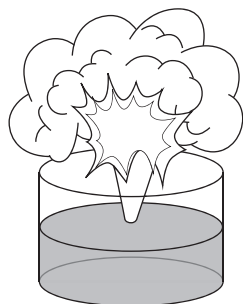
Tick (✓) the box that shows the melting point and boiling point values for potassium which fit the trends shown in the chart. [1]

melting point = 40 °C ☐  
boiling point = 600 °C

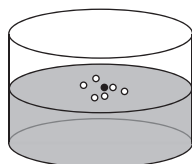
melting point = 60 °C ☐  
boiling point = 780 °C

melting point = 20 °C ☐  
boiling point = 600 °C

- (b) The diagrams show lithium, sodium, potassium and rubidium, **but not necessarily in that order**, reacting separately with cold water.

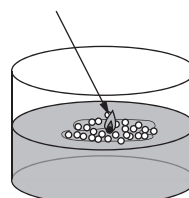
**A**

Violent reaction with sparks spitting out of the trough

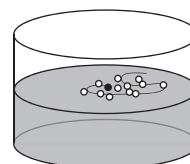
**B**

Gentle fizzing

lilac flame

**C**

Metal ball melts, moves around on the surface and burns

**D**

Metal ball fizzes and moves around on the surface

- (i) Use the information in the diagrams to give the **letter** which represents:

[2]

lithium .....

sodium .....

potassium .....

rubidium .....

- (ii) Caesium lies below rubidium in Group 1. Suggest how its reaction with water would be different from those above. [1]

.....

- (iii) Describe **one** safety precaution taken when adding a Group 1 metal to water. [1]

.....

- (c) Group 1 metals react with oxygen to form metal oxides.

Sodium oxide contains the ions  $\text{Na}^+$  and  $\text{O}^{2-}$ .

Underline the correct formula of sodium oxide.

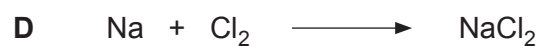
[1]



- (d) Group 1 metals react with chlorine,  $\text{Cl}_2$ , to form metal chlorides.

Give the **letter** for the balanced symbol equation for the reaction between sodium and chlorine.

[1]



Letter .....